Freight Rate Forecasting

Kipngeno Koech · Machine Learning Engineer assessment · Code: github.com/kkipngenokoech/freight-rate-forecasting

1. Problem framing

The labelled data (`train_test.csv`, 48,000 loads) covers **Jan–Oct 2025**. The 12,000 loads to predict cover **Nov–Dec 2025**. This is a two-month-ahead forecast, so a random train/test split would leak future market conditions into training and overstate accuracy. Every split below is by time.

2. Train/test split and validation approach

- **Primary holdout:** train on Jan–Aug, test on Sep–Oct. This is the same two-month horizon as the real task, and it includes a quarter end, as December does.
- **Rolling-origin backtest:** for each month from Jun to Oct, train on all earlier months and test on that month. This checks that the result is stable and not one lucky window.
- **Unseen-city check:** 8 cities in Nov–Dec never appear in training (1,447 loads). To mimic this, 8 cities are removed from the holdout's training set and those loads are scored separately.
- **Forecast-period conditions:** features that behave differently in Nov–Dec (see `quote_signal`, section 3) are scored with that behaviour simulated, not as they look in training.
- **Label noise:** about 1.4% of labels look corrupted. They are flagged using only each split's training window, so no holdout information leaks. Results are reported on clean rows and on all rows.

Primary holdout (train Jan–Aug, test Sep–Oct)	MAPE clean	MAE clean	MAPE all rows
Baseline: median $/mile by equipment × distance band	3.82%	$82.8	6.32%
Ridge regression on log $/mile	2.59%	$54.6	5.11%
Gradient boosting alone	3.65%	$86.8	6.06%
Ridge + gradient boosting hybrid (final)	**1.34%**	**$31.7**	**3.87%**

Rolling backtest, MAPE on clean rows	Jun	Jul	Aug	Sep	Oct	Mean
Ridge	2.52%	2.58%	2.58%	2.53%	2.65%	2.57%
Gradient boosting alone	2.31%	2.25%	3.05%	2.07%	5.13%	2.96%
Hybrid (final)	**1.27%**	**1.39%**	**1.32%**	**1.21%**	**1.48%**	**1.33%**

In the unseen-city check, loads touching a held-out city score **1.61%**, against 1.35% for the other loads.

3. Data findings and quality issues

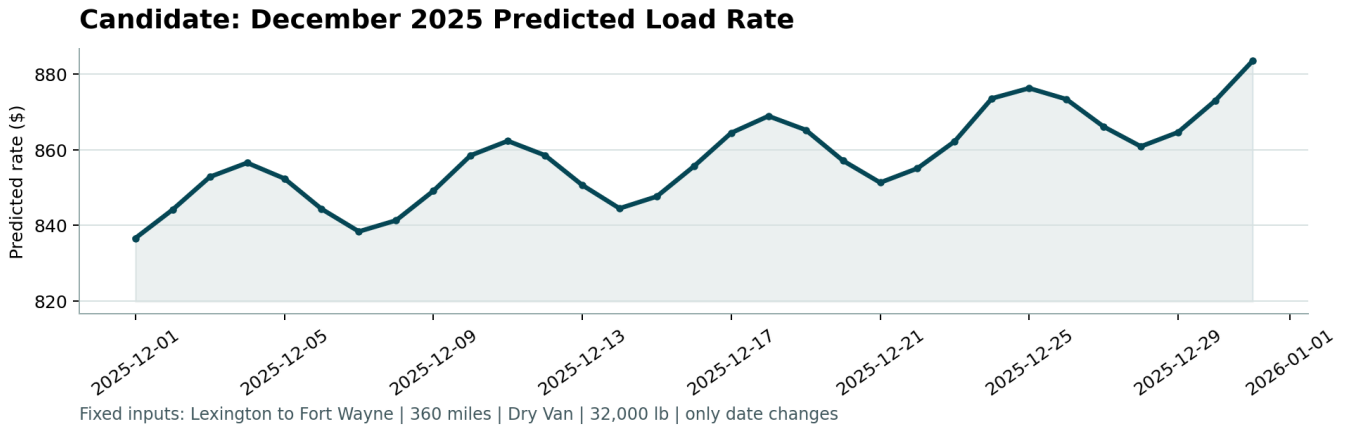
- **`quote_signal` leaks the label in training but is noise in the forecast period.** Across the whole dataset it looks useless (correlation 0.05). Within a single day it tracks $/mile closely (median |corr| 0.63), and the sign of that relationship flips every few weeks. The relationship is absent in August (0.05) and across all of Nov–Dec. A model that keeps it scores 1.35% on the holdout; with `quote_signal` shuffled within each day to mimic Nov–Dec, that rises to 1.51%. The feature is dropped.

- **Suspected corrupted labels (677 rows, 1.41%):** rates at 0.17–0.45× or 2.2–5.4× the expected rate, with an empty gap between them and normal rows. They are flagged with a first-pass model and excluded from training.
- **Negative weights** (292 train, 145 forecast): the sign is flipped, since these rows price like normal loads. **Weights capped at 47,500 lb** (1,191 rows): the value is kept and a cap flag is added. **Missing weight** (300) is left for the model to handle.
- **Missing `market_index`** (374): filled with that day's mean, because the index is a daily market level plus small noise.
- **Trend and seasonality:** rates drift up about +0.6% per month beyond what `market_index` explains, and they ramp up over the 5 weeks before each quarter end (most for flatbed, least for dry van).

4. Model

The model predicts log $/mile in two stages. A **ridge regression** captures effects that must extend past the training dates: distance curve, market index, equipment, weight, time trend, city effects, and a quarter-end ramp per equipment type. **Gradient boosting** then fits the ridge's residuals using geography, daily market level, day of week, days to quarter end and a shrunk per-lane effect. It gets no raw date, so the trend stays in the ridge. Boosting alone cannot extrapolate the trend: see its 5.13% in October. Loads touching a city not seen in training are routed to a twin model without city effects.

5. December prediction chart



Produced by the provided `score.py`. Both output files passed its validation (12,000 final predictions, 31 December rows). Lane: Lexington → Fort Wayne, 360 mi, Dry Van, 32,000 lb. The prediction follows the weekly market cycle and rises from about $837 to $883 into the quarter end. Historical rates on this lane ranged from $758 to $934 (mean $823, 21 loads).

6. Limitations

- The time trend is extended two months beyond the data.
- The month-to-month level, which jumps at each quarter start, is the largest remaining source of error.
- One year of data cannot capture holiday effects in late December.